Transposes, Permutations, Vector Spaces

Finish elimination with $PA = LU$, flip matrices with transposes, then step up from vectors to whole spaces of vectors, and the column space of a matrix.

PART 1 Index

What this lesson covers, in order. Click a topic to jump to it.

1. **Permutations: reordering rules**
   Re-listing the equations in another order: why a 0 forces it, PA = LU, and undoing it with P^T.
2. **Transposes**
   Rows become columns. Portfolio return as $r^T x$, and $(AB)^T = B^T A^T$ entry by entry.
3. **Symmetric matrices and R^TR**
   Why $R^T R$ is always symmetric, and how covariance matrices come from it.
4. **Vector spaces**
   A set you can add and scale inside without leaving: portfolios as a space.
5. **When a set is not a space**
   Long-only portfolios, fixed-budget portfolios and unions fail the test.
6. **Subspaces of R^2 and R^3**
   Lines and planes through the origin. Zero-cost (self-financing) trades form one.
7. **The column space C(A)**
   Everything the columns can build: which targets your funds can reach, one fund at a time.

METHODS IN THIS LESSON: PA = LU · $P^{-1} = P^T$ · Transpose · Dot product x^Ty · $(AB)^T = B^TA^T$ · R^TR is symmetric · Covariance matrix · Vector space test · Subspace test · Self-financing trades · Column space C(A)

PART 2 Prerequisites

Click any link to review it, then come back.

- Span and column space, first look (Lesson 1): "every point you can reach by mixing".
- Permutation matrices (Lesson 2): PA swaps rows, AP swaps columns.

- PA = LU and transposes of products (Lesson 4).
- The entry rule and building-block trades (Lesson 3).

PART 3 Explanation

THE RUNNING EXAMPLE: THE SAME FUNDS, NEW QUESTIONS

Funds: Nifty (12%, beta 1.0), bonds (7%, beta 0), midcap (15%, beta 1.2). Rules: budget, return (₹ thousand), market exposure. The fund matrix and the matrix of **scenario returns** we'll use:

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 12 & 7 & 15 \\ 1 & 0 & 1.2 \end{bmatrix} \qquad R = \begin{bmatrix} -10 & 4 \\ 10 & 7 \\ 30 & 10 \end{bmatrix} \begin{matrix} \leftarrow \text{bad year} \\ \leftarrow \text{normal year} \\ \leftarrow \text{good year} \end{matrix}$$

R lists the yearly return (%) of Nifty (column 1) and bonds (column 2) in three scenarios. This lesson asks three new questions. What happens when rows and columns swap roles? Which sets of portfolios behave like a "space"? And exactly which targets can a set of funds reach?

1. Permutations: changing the order of the rules

THE WHOLE IDEA IN ONE LINE

A **permutation** changes only the **order in which the equations are listed.** The equations themselves don't change, so **the answer doesn't change.** A permutation matrix P is just the instruction for the new order.

Step 1: two rules, two orders

Our two-fund problem has two rules. We can write them down in either order:

Order A (budget first)	Order B (return first)
(1) budget: $x + y = 10$ (2) return: $12x + 7y = 105$	(1) return: $12x + 7y = 105$ (2) budget: $x + y = 10$

It's the same two facts on a different line of the page, so both lists have the same answer: $x = 7$, $y = 3$ (check: $7 + 3 = 10$ and $84 + 21 = 105$). As matrices, the rows simply trade places:

$$\text{Order A: } \begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix} \quad \longrightarrow \quad \text{Order B: } \begin{bmatrix} 12 & 7 \\ 1 & 1 \end{bmatrix}$$

The matrix that makes this swap is $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. How to make it and how to read it:

1 MAKE IT

Do the same swap to the identity matrix $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$: swapping its two rows gives $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.

2 READ ROW 1

$(0, 1)$ means "take 0 of rule 1 and 1 of rule 2": $0 \times (1, 1) + 1 \times (12, 7) = (12, 7)$. The return rule now comes first.

READ ROW 2

$(1, 0)$ means "take rule 1": $1 \times (1, 1) + 0 \times (12, 7) = (1, 1)$. The budget rule now comes second.

3 DON'T FORGET B

The right-hand sides move with their rules: $Pb = P(10, 105) = (105, 10)$.

WHAT WE HAVE NOW

$PA = \begin{bmatrix} 12 & 7 \\ 1 & 1 \end{bmatrix}$ and $Pb = (105, 10)$: Order B, same answer $(7, 3)$.

Step 2: why would anyone reorder? A zero in the way

Take the hedging problem (sell x Nifty futures and y Nifty IT futures) and suppose someone wrote the currency rule first:

$$\begin{aligned} y &= 3 \quad \text{(currency)} \\ x + y &= 8 \quad \text{(market)} \end{aligned} \iff \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 8 \end{bmatrix}$$

1 ELIMINATION'S FIRST MOVE

Use the top-left number (the first **pivot**) to remove x from the row below. The multiplier is (number below) ÷ (pivot) $= 1 \div 0$.

PROBLEM

You can't divide by 0. The first rule has no x in it, so it can't be used to remove x from anything. Elimination is stuck.

2 FIX

Swap the two rules, using $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ again: market first, then currency.

WHAT WE HAVE NOW

$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 8 \\ 3 \end{bmatrix}$. It's already triangular: the last rule says $y = 3$, then the first says $x + 3 = 8$, so $x = 5$.

That is the only reason permutations show up in elimination: a 0 lands where a pivot should be, and swapping in a lower row that has a non-zero number there gets things moving again.

Step 3: three rules, six orders: try them

With three rules (market, currency, rates) there are 6 ways to list them. Writing **M**, **C**, **R** for the rules: M-C-R, M-R-C, C-M-R, C-R-M, R-M-C, R-C-M. Pick an order below. You'll see the rules in that order, the permutation matrix P that produces it, and whether elimination runs straight through or gets stuck on a 0.

M-C-R | M-R-C | C-M-R | C-R-M | R-M-C | R-C-M

1. **market:** $x + y + z = 10$
2. **currency:** $y = 3$
3. **rates:** $z = 2$

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$ (row i of P says which rule goes in place i)

Elimination runs straight through: pivots 1, 1, 1. Back substitution: z = 2, y = 3, x = 10 − 3 − 2 = 5.

Order	First rule listed	Elimination without swaps	Answer
M-C-R	market: $x + y + z = 10$	runs straight through (pivots 1, 1, 1)	$(5, 3, 2)$
M-R-C	market	stuck at pivot 2: the second rule (rates) has no y	$(5, 3, 2)$ after a swap
C-M-R, C-R-M	currency: $y = 3$	stuck at pivot 1: no x in the first rule	$(5, 3, 2)$ after a swap
R-M-C, R-C-M	rates: $z = 2$	stuck at pivot 1: no x in the first rule	$(5, 3, 2)$ after a swap

Two lessons from the table. **The answer is $(5, 3, 2)$ in every order**: reordering never changes it. And only **M-C-R** works without help, because the market rule is the only one containing x (Nifty futures), so it has to come first.

Step 4: PA = LU: reorder first, then eliminate

Elimination with a swap in the middle is untidy. Strang's cleaner way: **do all the reordering first**, then eliminate normally. The reordering is P; the normal elimination gives L and U (Lesson 4).

RULE · PA = LU

For every invertible A there is a permutation P that puts the rules in a workable order; then $PA = LU$ with no more swaps. If the order was already fine, $P = I$ and it's just $A = LU$.

One worked case: the credit fund. Lesson 2's credit fund earns 12% like Nifty but has beta 0. Rules in the usual order (budget, return, exposure):

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 12 & 12 & 15 \\ 1 & 0 & 1.2 \end{bmatrix}, \qquad b = \begin{bmatrix} 10 \\ 126 \\ 7.4 \end{bmatrix}$$

1 WHERE IT GETS STUCK

Return − 12 × budget gives $(12-12,\ 12-12,\ 15-12) = (0, 0, 3)$. Nifty **and** the credit fund disappear together (both earn 12), so row 2 has a 0 where the second pivot should be.

THE FIX

List the exposure rule second instead: order budget, exposure, return. That's P_{23} (swap rows 2 and 3 of I): $P_{23}A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1.2 \\ 12 & 12 & 15 \end{bmatrix}$.

2 NOW ELIMINATE

Exposure − 1 × budget: $(1-1,\ 0-1,\ 1.2-1) = (0, -1, 0.2)$. Return − 12 × budget: $(0, 0, 3)$. The entry below the second pivot is already 0. Done: $U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -1 & 0.2 \\ 0 & 0 & 3 \end{bmatrix}$, and the multipliers 1, 12, 0 give $L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 12 & 0 & 1 \end{bmatrix}$.

3 CHECK LU = PA

Row 2 of LU: $1 \times (1,1,1) + 1 \times (0,-1,0.2) = (1, 0, 1.2)$ ✓. Row 3: $12 \times (1,1,1) + 1 \times (0,0,3) = (12, 12, 15)$ ✓. (Row 1 is just $(1,1,1)$ ✓.)

4 SOLVE

Reorder b the same way: $Pb = (10, 7.4, 126)$. Forward: $c = (10,\ 7.4-10,\ 126-120) = (10, -2.6, 6)$. Back: $3z = 6$ so $z = 2$; $-y + 0.4 = -2.6$ so $y = 3$; $x + 3 + 2 = 10$ so $x = 5$.

ANSWER

₹5 lakh Nifty, ₹3 lakh credit, ₹2 lakh midcap. Check: 10 ✓, $60 + 36 + 30 = 126$ ✓, $5 + 2.4 = 7.4$ ✓.

Step 5: undoing a reorder: $P^{-1} = P^T$

To undo a reorder, put the rules back. Two easy cases:

- **A swap:** swapping twice puts everything back, so $P \times P = I$. Check for $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$: rows of PP are $0(0,1) + 1(1,0) = (1,0)$ and $1(0,1) + 0(1,0) = (0,1)$, which is I ✓. A swap is its own undo.
- **A cycle:** P turns the order (M, C, R) into (C, R, M). To undo it, turn (C, R, M) back into (M, C, R). The matrix that does that turns out to be the transpose P^T.

> **RULE · $P^{-1} = P^T$**
> The undo of any reorder is its transpose: $P^T P = I$. So you never need to "compute" the inverse of a permutation: just flip it.

Check for the cycle. $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ (take rule 2, rule 3, rule 1) and $P^T = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$. Multiply $P^T P$ row by row: row 1 of P^T is $(0,0,1)$, "take row 3 of P" $= (1,0,0)$; row 2 is $(1,0,0)$, "take row 1 of P" $= (0,1,0)$; row 3 is $(0,1,0)$, "take row 2 of P" $= (0,0,1)$. That is I ✓.

How many orders are there? 2 rules: 2 orders. 3 rules: $3 \times 2 \times 1 = 6$ (any of 3 first, either of the other 2 next, the last one is forced). n rules: $n! = n(n-1)\cdots 1$, which is 24 for 4 rules and 120 for 5.

▼ For the curious: small pivots, and permutations as a "family"

Why software swaps more than it has to. Algebra only needs a swap when a pivot is exactly 0. Programs like MATLAB also swap when a pivot is merely *small*, because a tiny pivot means a huge multiplier, which magnifies rounding errors. Example: $0.0001x + y = 1$ and $x + y = 2$ (true answer $x \approx 1.0001$, $y \approx 0.9999$). With pivot 0.0001 the multiplier is 10,000; if y is rounded to 1.000, then $x = (1 - y)/0.0001 = 0$, completely wrong. Swap first and the pivot is 1: the same rounded y gives $x = 2 - y = 1.000$, correct. Picking the largest available pivot is called **partial pivoting**.

A closed family. Reordering twice is still a reordering, so multiplying two permutation matrices gives another one, and every one has an undo. For example, swap rows 2 and 3, then rows 1 and 2: $P_{12}P_{23} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$, a cycle. Mathematicians call such a family a *group*.

> **IN ONE LINE**
> A permutation only re-lists the rules (and their right-hand sides); it's needed when a 0 sits where a pivot should be; $PA = LU$ means "reorder first, then eliminate"; and the undo is P^T.

▼ Check yourself (MIT problem 5.1): find a 3 × 3 permutation P, not the identity, with $P^3 = I$.

The cycle (M, C, R) → (C, R, M). Apply it three times: (C, R, M) → (R, M, C) → (M, C, R), back to the start. So $P^3 = I$.

2. Transposes

METHOD · TRANSPOSE

The **transpose** A^T flips A across its main diagonal: rows become columns and columns become rows. In symbols, $(A^T)_{ij} = A_{ji}$: the entry in row i, column j of A^T is the entry in row j, column i of A. An $m \times n$ matrix has an $n \times m$ transpose.

Transposing the fund matrices

1 2 × 2

$A = \begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix}$. Row 1 $(1, 1)$ becomes column 1; row 2 $(12, 7)$ becomes column 2: $A^T = \begin{bmatrix} 1 & 12 \\ 1 & 7 \end{bmatrix}$.

CHECK ONE ENTRY

$(A^T)_{12}$ should be $A_{21} = 12$ ✓.

2 3 × 3

RESULT

Each row of A becomes a column:
$A^T = \begin{bmatrix} 1 & 12 & 1 \\ 1 & 7 & 0 \\ 1 & 15 & 1.2 \end{bmatrix}$. Diagonal entries $1, 7, 1.2$ stay put; $A_{21} = 12$ moved to position $(1, 2)$, $A_{32} = 0$ to $(2, 3)$, and so on.

3 RECTANGULAR

With only Nifty and bonds but all three rules, A is 3×2: $\begin{bmatrix} 1 & 1 \\ 12 & 7 \\ 1 & 0 \end{bmatrix}$. Its transpose is 2×3: $\begin{bmatrix} 1 & 12 & 1 \\ 1 & 7 & 0 \end{bmatrix}$.

MEANING

In A each **column** is a fund; in A^T each **row** is a fund's full profile: Nifty $= (1, 12, 1)$, bonds $= (1, 7, 0)$.

Row vectors and the dot product

Vectors are columns. Their transposes are rows. That gives a compact way to write a dot product:

METHOD · DOT PRODUCT AS X^TY

For column vectors x and y with n entries, $x^T y$ is a $(1 \times n)(n \times 1) = 1 \times 1$ product: a single number, $x_1 y_1 + \cdots + x_n y_n$.

Portfolio return in one line. Let $r = (12, 7, 15)$ be the returns per lakh and $x = (5, 3, 2)$ the holdings. Then

$$r^T x = \begin{bmatrix} 12 & 7 & 15 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \\ 2 \end{bmatrix} = 12 \times 5 + 7 \times 3 + 15 \times 2 = 60 + 21 + 30 = 111.$$

₹111 thousand: exactly the return rule (row 2 of A) applied to client 1. Finance writes portfolio return as $w^T r$ ("weights transpose returns") all the time; this is that formula.

The transpose of a product reverses the order

RULE · $(AB)^T = B^T A^T$

Transpose each factor and reverse the order, just like inverses (Lesson 4). Also $(A^T)^T = A$: flipping twice returns the original.

Check it on Lesson 3's two clients, $A = \begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix}$ and $X = \begin{bmatrix} 7 & 4 \\ 3 & 6 \end{bmatrix}$, where $AX = \begin{bmatrix} 10 & 10 \\ 105 & 90 \end{bmatrix}$:

1 LEFT SIDE

Transpose AX: $(AX)^T = \begin{bmatrix} 10 & 105 \\ 10 & 90 \end{bmatrix}$.

2 RIGHT SIDE

$X^T = \begin{bmatrix} 7 & 3 \\ 4 & 6 \end{bmatrix}$ and $A^T = \begin{bmatrix} 1 & 12 \\ 1 & 7 \end{bmatrix}$. Entry by entry: $(7 \cdot 1 + 3 \cdot 1,\ 7 \cdot 12 + 3 \cdot 7) = (10, 105)$ for row 1, and $(4 \cdot 1 + 6 \cdot 1,\ 4 \cdot 12 + 6 \cdot 7) = (10, 90)$ for row 2.

WHAT WE HAVE NOW

$X^T A^T = \begin{bmatrix} 10 & 105 \\ 10 & 90 \end{bmatrix} = (AX)^T$ ✓.

3 MEANING

AX had one **column** per client; $(AX)^T$ has one **row** per client: (budget, return) for client 1, then client 2. That's the same report sorted the other way, like a spreadsheet with clients as rows.

WHY NOT $A^T X^T$?

$A^T X^T = \begin{bmatrix} 1{\cdot}7{+}12{\cdot}4 & 1{\cdot}3{+}12{\cdot}6 \\ 1{\cdot}7{+}7{\cdot}4 & 1{\cdot}3{+}7{\cdot}6 \end{bmatrix} = \begin{bmatrix} 55 & 75 \\ 35 & 45 \end{bmatrix}$: wrong order, meaningless numbers.

IN ONE LINE

$(A^T)_{ij} = A_{ji}$; $x^T y$ is a dot product; $(AB)^T = B^T A^T$.

▼ Check yourself (Strang's example): transpose $\begin{bmatrix}1&3\\2&3\\4&1\end{bmatrix}$.

Column 1 $(1,2,4)$ becomes row 1 and column 2 $(3,3,1)$ becomes row 2: $\begin{bmatrix}1&2&4\\3&3&1\end{bmatrix}$, a 2×3 matrix.

3. Symmetric matrices and RᵀR

DEFINITION: SYMMETRIC MATRIX

A square matrix is **symmetric** if transposing doesn't change it: $S^T = S$, that is $S_{ij} = S_{ji}$ for every i, j. Everything above the diagonal is mirrored below it. Example: $\begin{bmatrix}3&1&7\\1&2&9\\7&9&4\end{bmatrix}$.

Strang: symmetric matrices "show up in a lot of applications", and he shows where they come from. Take **any** matrix R, even a rectangular one, and multiply its transpose by it.

RULE · RᵀR IS ALWAYS SYMMETRIC

For any $m \times n$ matrix R, the $n \times n$ matrix $R^T R$ is symmetric. Proof in two lines: $(R^T R)^T = R^T (R^T)^T = R^T R$, using "reverse the order" and $(R^T)^T = R$. (RR^T, which is $m \times m$, is symmetric for the same reason.)

RᵀR for the scenario returns, entry by entry

R is 3×2 (3 scenarios × 2 funds), so $R^T R$ is 2×2. Entry (i, j) = (column i of R) · (column j of R). The columns are Nifty $(-10, 10, 30)$ and bonds $(4, 7, 10)$.

1 ENTRY (1, 1)

Nifty · Nifty $= (-10)^2 + 10^2 + 30^2 = 100 + 100 + 900 = 1100.$

2 ENTRY (1, 2)

Nifty · bonds $= (-10)(4) + (10)(7) + (30)(10) = -40 + 70 + 300 = 330.$

ENTRY (2, 1)

bonds · Nifty: the same three products in the other order, $4(-10) + 7(10) + 10(30) = 330$. **This is the symmetry:** a dot product doesn't care which vector comes first.

3 ENTRY (2, 2)

bonds · bonds $= 4^2 + 7^2 + 10^2 = 16 + 49 + 100 = 165.$

WHAT WE HAVE NOW

$R^T R = \begin{bmatrix}1100&330\\330&165\end{bmatrix}$: symmetric ✓.

The 3 × 3 version RR^T compares **scenarios** instead of funds (entry (i, j) = row i · row j of R):

$$RR^T = \begin{bmatrix} (-10)^2 + 4^2 & (-10)(10) + 4 \cdot 7 & (-10)(30) + 4 \cdot 10 \\ \cdot & 10^2 + 7^2 & 10 \cdot 30 + 7 \cdot 10 \\ \cdot & \cdot & 30^2 + 10^2 \end{bmatrix} = \begin{bmatrix} 116 & -72 & -260 \\ -72 & 149 & 370 \\ -260 & 370 & 1000 \end{bmatrix}$$

Only the upper triangle needed computing; symmetry fills in the rest.

MODEL · COVARIANCE MATRIX (PREVIEW)

Subtract each fund's average return first, then form $\frac{1}{m} C^T C$ (with m scenarios): that is the **covariance matrix** of the funds, the core input of portfolio risk models. Because it has the form $C^T C$, **a covariance matrix is always symmetric**.

1 AVERAGES

Nifty: $(-10 + 10 + 30)/3 = 10$. Bonds: $(4 + 7 + 10)/3 = 7$.

SUBTRACT THEM

Nifty: $(-10 - 10,\ 10 - 10,\ 30 - 10) = (-20, 0, 20)$. Bonds: $(4 - 7,\ 7 - 7,\ 10 - 7) = (-3, 0, 3)$. These are the columns of C.

2 $C^T C$

$(1, 1)$: $400 + 0 + 400 = 800$. $(1, 2) = (2, 1)$: $(-20)(-3) + 0 + 20 \cdot 3 = 60 + 60 = 120$. $(2, 2)$: $9 + 0 + 9 = 18$.

DIVIDE BY 3

$$\text{Covariance} = \begin{bmatrix} 800/3 & 40 \\ 40 & 6 \end{bmatrix} \approx \begin{bmatrix} 266.7 & 40 \\ 40 & 6 \end{bmatrix}.$$

MEANING

Nifty's variance ≈ 266.7 (volatility ≈ $\sqrt{266.7} \approx 16.3\%$), bonds' variance 6 (≈ 2.4%), and a positive covariance 40: in these scenarios the two funds move together. Risk models come back in the probability course.

IN ONE LINE

$S^T = S$ means mirrored across the diagonal; $R^T R$ is always symmetric, which is why covariance matrices are.

▼ Check yourself (MIT problem 5.2): how many entries of a 4 × 4 matrix can be chosen freely if it is symmetric? If it is skew-symmetric ($A^T = -A$)?

Symmetric: the 4 diagonal entries plus the 6 above the diagonal (the 6 below are copies): **10**. Skew-symmetric: the diagonal must be 0 (since $a_{ii} = -a_{ii}$) and the lower triangle is minus the upper: **6**.

4. Vector spaces

> THE WHOLE IDEA IN ONE LINE
>
> A **vector space** is a set of vectors you **can't escape by adding or scaling**: whatever you build from members of the set, by adding them or multiplying them by numbers, is again a member.

Strang calls this "the beginning of real linear algebra". Until now we looked at single vectors (one portfolio, one target). From here on we look at **whole collections** of vectors, and ask whether a collection is "closed". Here's what that means, step by step.

Step 1: the two moves

There are only two things we ever do with vectors. With portfolios (x, y) = (₹ lakh in Nifty, ₹ lakh in bonds):

Move	How	Example	In portfolio terms
Add two vectors	add matching components	$(7,3)+(4,6) = (7+4,\ 3+6) = (11,9)$	merge two clients' books into one
Scale a vector by a number c	multiply every component by c	$2 \times (7,3) = (14,6)$	double the position
		$0.5 \times (7,3) = (3.5, 1.5)$	halve it
		$-1 \times (7,3) = (-7,-3)$	flip it: sell short instead of buy
		$0 \times (7,3) = (0,0)$	close everything: the empty portfolio

Doing both at once, "scale some vectors and add", is a **linear combination** (Lesson 1), like $2(7,3)+1(4,6) = (18,12)$.

Step 2: a warm-up on the number line

Before vectors, think of plain numbers, and ask: "if I start inside this set, can adding or multiplying take me outside it?"

1 ALL REAL NUMBERS

Add any two: $3 + 5 = 8$, a real number. Multiply by any number: $-2 \times 3 = -6$, a real number. You can never leave.

VERDICT

Closed: this is the simplest vector space, $\mathbb{R}$ (the number line).

2 POSITIVE NUMBERS ONLY

Adding is fine: $3 + 5 = 8$ is positive. But multiplying by -1: $-1 \times 3 = -3$, **not positive**. You escaped.

VERDICT

Not closed: one escape is enough to fail.

That is the entire test. For vectors it's exactly the same, just with pairs or triples of numbers.

RULE · VECTOR SPACE TEST (CLOSURE)

A set V of vectors is a **vector space** when both moves keep you inside, for every choice:

1. **Adding:** if v and w are in V, then $v + w$ is in V.
2. **Scaling:** if v is in V and c is *any* real number (positive, negative or zero), then $c\,v$ is in V.

Together these say every linear combination $c\,v + d\,w$ stays in V. (Textbooks also list eight "good behaviour" rules, such as $v + w = w + v$. For lists of numbers they always hold, so in practice only the two above need checking.)

Step 3: the portfolio plane passes

Take the set of **all** portfolios (x, y) of Nifty and bonds, long or short, any real amounts. That is $\mathbb{R}^2$, the whole xy-plane.

1 ADDING

$(7, 3) + (4, 6) = (11, 9)$. Adding two real numbers always gives a real number, so the sum of any two pairs is again a pair of real numbers ✓.

2 SCALING

$2(7, 3) = (14, 6)$, $-1(7, 3) = (-7, -3)$, $0(7, 3) = (0, 0)$: all pairs of real numbers ✓. Shorting is allowed here, so negative amounts are fine.

VERDICT

Nothing ever escapes: $\mathbb{R}^2$ **is a vector space.**

Step 4: zero is always inside: a quick first test

Scaling by $c = 0$ is always allowed. So if v is in a vector space, $0 \times v = (0, 0)$ must be too. That gives a very fast check:

> THE ZERO TEST
>
> **Every vector space contains the zero vector.** If a set doesn't contain 0, it fails immediately, with no other checking needed. (Strang: "Every vector space has got that zero vector in it.") For example, "all portfolios with a ₹10 lakh budget" doesn't contain $(0, 0)$, whose budget is 0, so it is not a vector space. The plane with the origin removed fails too.

The converse doesn't hold: containing 0 is necessary, not enough. Long-only portfolios contain $(0, 0)$ but still fail (topic 5).

Step 5: more funds, more components: R^3 and R^n

$\mathbb{R}^3$ is all vectors with 3 real components, such as three-fund portfolios $(5, 3, 2)$ = (Nifty, bonds, midcap). $\mathbb{R}^n$ is all vectors with n components: a portfolio of n funds. Each passes the test for the same reason $\mathbb{R}^2$ does: adding and scaling real numbers gives real numbers.

Careful: $(3, 2, 0)$ is in $\mathbb{R}^3$, not $\mathbb{R}^2$. It has three components (no midcap, but the slot exists). The number of components decides which space a vector lives in.

Try it: the space tester

Pick a set of portfolios, two members v and w, and a number c. The tester computes $v + w$ and $c\,v$ and tells you whether each stayed inside. Try to make the "not a space" sets fail; try to make "all portfolios" fail (you can't).

Set [All portfolios (R²) ▾]

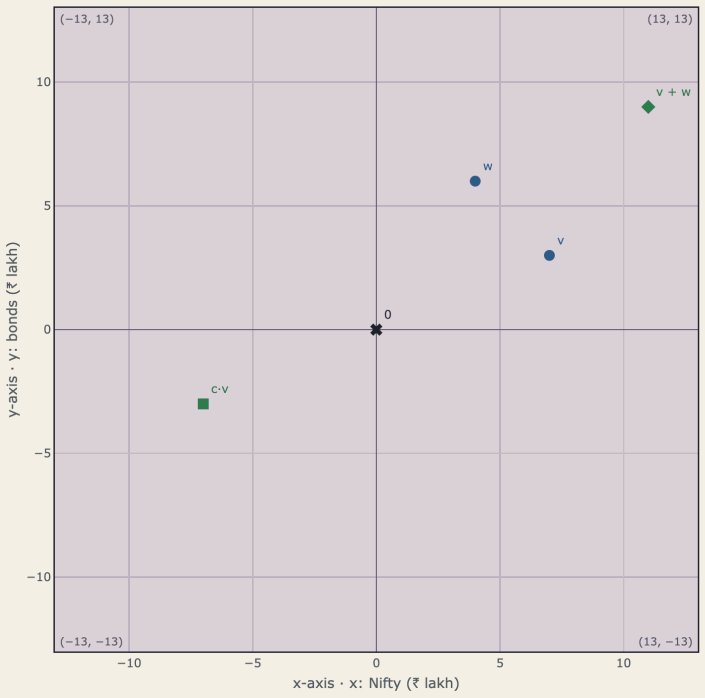
v = (7 , 3) w = (4 , 6) c = -1

Zero vector $(0, 0)$: **inside ✓**

Add: $v + w = (7, 3) + (4, 6) = (11, 9) \rightarrow$ **inside ✓**

Scale: $c\,v = -1 \times (7, 3) = (-7, -3) \rightarrow$ **inside ✓**

No escape with these numbers. A vector space must pass for *every* choice, so try others: all portfolios, zero-cost trades and all-Nifty never fail.



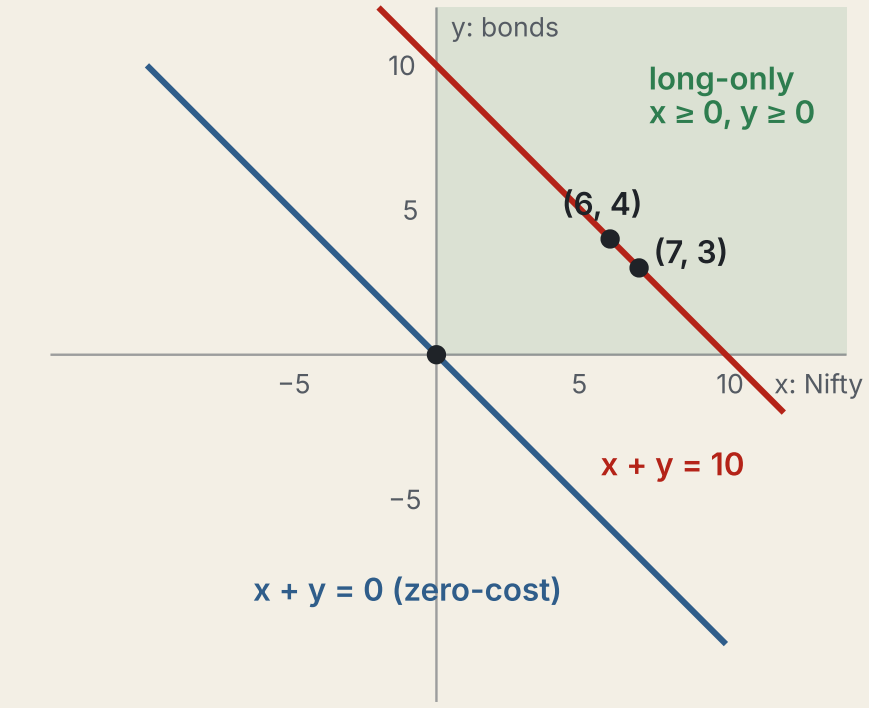
Shaded or drawn in purple: the set. Blue: v and w. Green or red: $v + w$ and $c\,v$ (red means it escaped).

IN ONE LINE

A vector space is closed under adding and scaling (so under every linear combination); it always contains 0, and one escape is enough to fail.

5. When a set is not a space

The test is easiest to understand from sets that fail it. Here are three, all natural in portfolio work.



The portfolio plane. Shaded: long-only portfolios. Red: all portfolios with a ₹10 lakh budget. Blue: zero-cost trades, $x + y = 0$. Only the blue line passes the vector-space test.

Failure 1: long-only portfolios (a quadrant)

Many funds may only **buy**, never sell short: $x \geq 0$ and $y \geq 0$. Strang uses exactly this quarter-plane as his first non-example.

1	ADDITION	Two long-only portfolios, $(7, 3) + (6, 4) = (13, 7)$: both components still ≥ 0 ✓. Adding is fine.
2	SCALING	Multiply $(7, 3)$ by -1: $(-7, -3)$, both negative, so outside the quadrant ✗. (Strang multiplies by -5; any negative number fails.)
	VERDICT	Closed under addition, **not** under scalar multiplication: **not a vector space.**

IN MARKETS

This is why a **long-only** constraint changes the maths of portfolio construction. Without it, the set of allowed portfolios is a vector space and linear algebra solves everything directly. With it, you need inequality-constrained optimisation, a different toolkit.

Failure 2: a fixed budget (a line not through the origin)

All portfolios that use exactly ₹10 lakh: $x + y = 10$ (the red line).

1	ADDITION	$(7, 3)$ and $(6, 4)$ both have budget 10. Their sum $(13, 7)$ has budget $13 + 7 = 20 \neq 10$ ✗.
2	ZERO	$0 \times (7, 3) = (0, 0)$ has budget $0 \neq 10$ ✗. The zero vector is not on the line.
	VERDICT	**Not a vector space.** Strang: a line in $\mathbb{R}^2$ that misses the origin is never a subspace.

Failure 3: "all-Nifty or all-bonds" (a union of two lines)

From the recitation: take every all-Nifty portfolio $(t, 0)$ together with every all-bond portfolio $(0, t)$.

1	SCALING	A multiple of an all-Nifty portfolio is still all-Nifty, and the same for bonds ✓.
2	ADDITION	$(1, 0) + (0, 1) = (1, 1)$: a mixed portfolio, on neither line ✗.
	VERDICT	**Not a vector space.** Each line alone is a subspace, but their union isn't: adding escapes it.

IN ONE LINE

To fail, a set only has to let one sum or one multiple escape: long-only (negatives), fixed budget (sums and 0), unions (sums).

6. Subspaces of R² and R³

RULE · SUBSPACE TEST

A **subspace** is a vector space sitting inside a bigger one. To check that a set inside $\mathbb{R}^n$ is a subspace: (1) it contains 0; (2) sums stay inside; (3) multiples stay inside. Strang: once a subspace contains a non-zero vector v, it must contain the **whole line** of multiples $c\,v$.

All the subspaces of R^2

Subspace	Example in portfolio terms
All of $\mathbb{R}^2$	every portfolio of Nifty and bonds, long or short
A line through $(0, 0)$	all-Nifty portfolios $(t, 0)$; or zero-cost trades $(t, -t)$
Just $\{(0, 0)\}$	the empty portfolio alone

There is nothing in between. A line through the origin inside $\mathbb{R}^2$ looks like $\mathbb{R}^1$ but isn't the same thing: its vectors still have **two** components (Strang's point).

Zero-cost trades: a subspace that matters

> **MODEL · SELF-FINANCING (ZERO-COST) TRADES**
>
> A trade is **self-financing** (zero-cost) if the money it puts into some funds is exactly taken out of others: the changes add up to zero. Rebalancing an existing portfolio without adding or withdrawing money is always a zero-cost trade.

With two funds the zero-cost trades are $\{(x, y) : x + y = 0\}$, the blue line through $(1, -1)$. Run the subspace test:

1 ZERO — $(0, 0)$: $0 + 0 = 0$ ✓ ("do nothing" costs nothing).

2 SUMS — Two zero-cost trades $(a, -a)$ and $(b, -b)$ add to $(a + b,\ -a - b) = (a + b,\ -(a + b))$: components still add to 0 ✓. Example: $(2, -2) + (1, -1) = (3, -3)$.

3 MULTIPLES — $c\,(a, -a) = (ca, -ca)$: still adds to 0 ✓. Example: $-3 \times (2, -2) = (-6, 6)$.

VERDICT — **A subspace**: a line through the origin. Compare with "budget = 10": the only difference is the right-hand side, 0 instead of 10, and that 0 is what lets the origin in.

All the subspaces of $\mathbb{R}^3$

Subspace	Example with three funds
All of $\mathbb{R}^3$	every three-fund portfolio
A plane through 0	zero-cost trades: $x + y + z = 0$
A line through 0	zero-cost trades that also keep market exposure unchanged
Just $\{0\}$	the empty trade

Working out that line. A trade (x, y, z) that changes neither the budget nor the exposure needs $x + y + z = 0$ and $x + 1.2z = 0$.

1 FROM THE EXPOSURE RULE $x = -1.2z.$

FROM THE BUDGET RULE $y = -x - z = 1.2z - z = 0.2z.$

2 SO every such trade is $z \times (-1.2, 0.2, 1)$. Taking $z = 5$ gives whole numbers: $(-6, 1, 5)$.

CHECK Budget: $-6 + 1 + 5 = 0$ ✓. Exposure: $-6 + 1.2 \times 5 = -6 + 6 = 0$ ✓. Return: $12(-6) + 7(1) + 15(5) = -72 + 7 + 75 = 10$.

MEANING The trades that change **only the return** form a line, and this one adds ₹10 thousand. It is exactly $10\times$ the building-block column $(-0.6, 0.1, 0.5)$ of A^{-1} in Lesson 3.

That line is where the zero-cost plane and the zero-exposure plane meet. In general, **the intersection of two subspaces is a subspace** (both conditions survive), while a union usually isn't (topic 5). Add a third condition, "zero return change", and only $(0, 0, 0)$ is left: the three planes meet only at the origin, because the fund matrix is invertible.

> **IN ONE LINE**
>
> Subspaces are $\{0\}$, lines and planes through the origin, and the whole space. Zero-cost trades are a subspace; fixed-budget portfolios are not.

▼ Check yourself (MIT problem 5.3): do the symmetric 2 × 2 matrices form a subspace of all 2 × 2 matrices? Do the non-symmetric ones?

Symmetric: yes. If $A^T = A$ and $B^T = B$ then $(A+B)^T = A^T + B^T = A + B$ and $(cA)^T = cA^T = cA$; and the zero matrix is symmetric. Non-symmetric: no. $\begin{bmatrix}1 & 1\\ 0 & 0\end{bmatrix} + \begin{bmatrix}0 & 0\\ 1 & 1\end{bmatrix} = \begin{bmatrix}1 & 1\\ 1 & 1\end{bmatrix}$, which is symmetric, so the sum escapes (and 0 isn't in the set at all).

7. The column space C(A)

THE WHOLE IDEA IN ONE LINE

The **column space** $C(A)$ is **everything you can build by mixing the columns of** A in any amounts. With funds as columns, it is exactly **the set of targets your funds can reach.**

THE KEY DIFFERENCE

Vector space: "What is the entire playground of vectors?"

Column space: "What playground can I create using these particular columns?"

Every lesson so far asked "which portfolio hits this target?". The column space asks the question before that one: **which targets can be hit at all?** We'll build the answer one fund at a time.

METHOD · COLUMN SPACE C(A)

For a matrix A with m rows, $C(A)$ is the set of all linear combinations of its columns: every Ax, over all possible x. It always lives in $\mathbb{R}^m$ (one component per rule).

Step 1: one fund only: a line

Suppose you may only use Nifty. One ₹ lakh of Nifty does $(1, 12, 1)$ to the rules (budget, return, exposure). Holding t lakh does $t \times (1, 12, 1)$:

₹ lakh in Nifty (t)	Target reached $t \times (1, 12, 1)$
1	$(1, 12, 1)$
5	$(5, 60, 5)$
-2 (short)	$(-2, -24, -2)$
0	$(0, 0, 0)$

All of these are multiples of one arrow, so they lie on **a line through the origin**. With one fund you can only reach targets on that line, like "₹5 lakh, ₹60 thousand, exposure 5", and nothing off it.

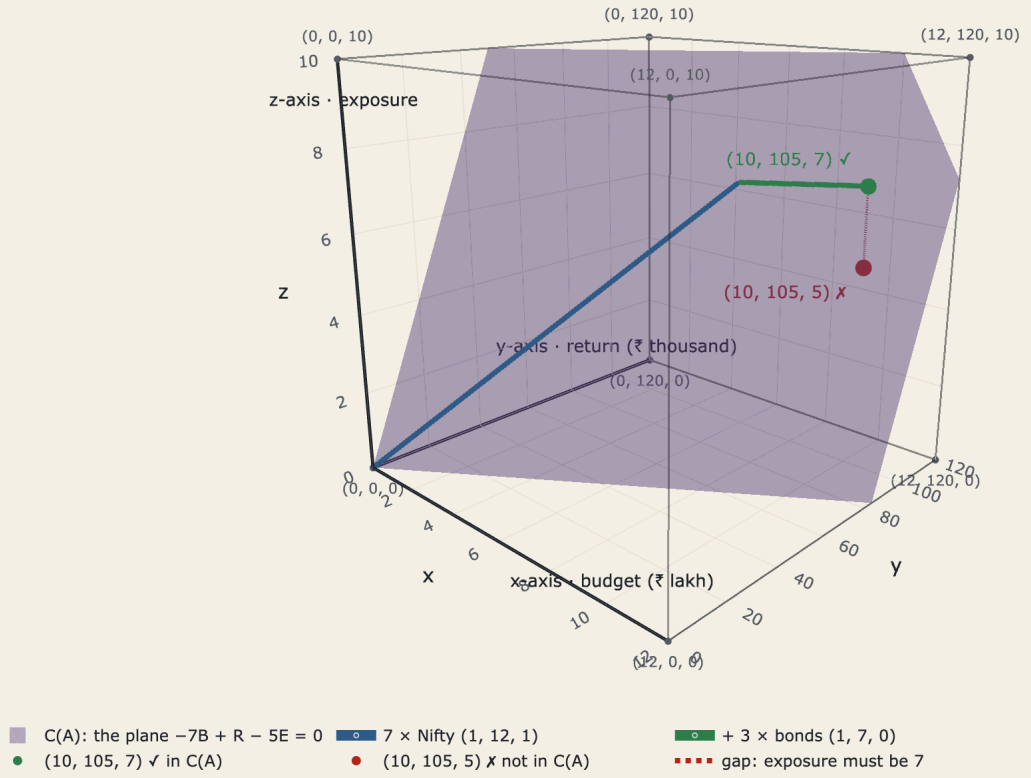
Step 2: two funds: a plane

Now allow Nifty **and** bonds, still with all three rules. The fund matrix has two columns:

$$A = \begin{bmatrix} 1 & 1 \\ 12 & 7 \\ 1 & 0 \end{bmatrix} \begin{matrix} \leftarrow \text{budget} \\ \leftarrow \text{return} \\ \leftarrow \text{exposure} \end{matrix} \qquad x \text{ lakh Nifty} + y \text{ lakh bonds reaches } x \begin{bmatrix} 1 \\ 12 \\ 1 \end{bmatrix} + y \begin{bmatrix} 1 \\ 7 \\ 0 \end{bmatrix}.$$

Mix (x, y)	Target reached	Arithmetic
$(1, 1)$	$(2, 19, 1)$	$(1+1,\ 12+7,\ 1+0)$
$(7, 3)$	$(10, 105, 7)$	$(7+3,\ 84+21,\ 7+0)$
$(3, -1)$	$(2, 29, 3)$	$(3-1,\ 36-7,\ 3-0)$

The bond arrow points in a different direction from the Nifty arrow, so mixing them sweeps out more than a line: a flat **plane through the origin**. But two directions can't fill all of 3D space, so it is **only** a plane.



$C(A)$ for Nifty + bonds: a plane through the origin (drag to rotate). 7 Nifty steps plus 3 bond steps reach $(10, 105, 7)$ on the plane. The target $(10, 105, 5)$ floats off it.

Step 3: why only a plane: the return is forced

Here is the plain reason. With only Nifty (beta 1) and bonds (beta 0):

1 EXPOSURE

Only Nifty adds exposure, 1 per lakh, so **exposure E = the Nifty amount x.**

BUDGET

$B = x + y$, so the bond amount is $y = B - E$.

2 RETURN

$R = 12x + 7y = 12E + 7(B - E) = 12E + 7B - 7E = 7B + 5E$.

WHAT IT MEANS

Once you choose the budget and the exposure, **the return is already decided:** it has to be $7B + 5E$. You can't pick all three freely.

THE REACHABLE TARGETS

With Nifty and bonds, a target (B, R, E) is reachable **exactly when** $R = 7B + 5E$. That single condition is the plane. (Written as $-7B + R - 5E = 0$, it's the same plane drawn above.)

Check it on the mixes from step 2: $(2, 19, 1)$: $7 \times 2 + 5 \times 1 = 19$ ✓. $(10, 105, 7)$: $70 + 35 = 105$ ✓. $(2, 29, 3)$: $14 + 15 = 29$ ✓. Every one obeys the rule.

Step 4: is my target reachable?

1

TARGET — $(10, 105, 7)$: ₹10 lakh, ₹105 thousand, exposure 7.

TEST — Forced return $= 7 \times 10 + 5 \times 7 = 70 + 35 = 105$. Wanted: 105. They match ✓.

THE MIX — $x = E = 7, y = B - E = 3$: ₹7 lakh Nifty, ₹3 lakh bonds. **In** $C(A)$.

2

TARGET — $(10, 105, 5)$: the same budget and return, but exposure only 5.

TEST — Forced return $= 70 + 5 \times 5 = 95$. Wanted: 105. Short by 10 ✗.

VERDICT — **Not in** $C(A)$. With exposure 5 you must hold ₹5 lakh Nifty and ₹5 lakh bonds, and that earns only ₹95 thousand. No mix of these two funds does better.

Step 5: a third fund: does it add a new direction?

Add a third column. The test is simple: **does the new fund obey** $R = 7B + 5E$ (then it's already on the plane and adds nothing), or not (then it points off the plane and the reach grows to all of $\mathbb{R}^3$)?

Third fund	Column (B, R, E)	$7B + 5E$	On the plane?	$C(A)$ becomes
Midcap	$(1, 15, 1.2)$	$7 + 6 = 13 \neq 15$	no: a new direction	**all of** $\mathbb{R}^3$: every target reachable
Balanced (½ Nifty + ½ bonds)	$(1, 9.5, 0.5)$	$7 + 2.5 = 9.5$	yes: nothing new	still the same plane

That's why every client in Lessons 1–4 had an answer (the midcap fund made $C(A)$ the whole space), and why the balanced fund was "singular" in Lesson 1: it lay in the plane the other two

already covered.

Step 6: why C(A) is always a subspace

Run topic 4's test on "all mixes of the columns". Adding two mixes, $(x_1N + y_1B) + (x_2N + y_2B) = (x_1 + x_2)N + (y_1 + y_2)B$, is again a mix ✓. Scaling a mix, $c(xN + yB) = (cx)N + (cy)B$, is again a mix ✓. And the "zero mix" $0N + 0B = 0$ is inside ✓. So $C(A)$ is always a subspace: a point, a line, a plane, or the whole space.

Columns pointing in…	$C(A)$ in $\mathbb{R}^3$ is	Example
1 direction	a line	Nifty only (or two identical Nifty funds)
2 directions	a plane	Nifty + bonds; Nifty + bonds + balanced
3 directions	all of $\mathbb{R}^3$	Nifty + bonds + midcap

WHAT'S NEXT

$Ax = b$ has a solution **exactly when b is in $C(A)$**: "is there a portfolio for this target?" is the same as "is the target reachable?". The next lesson builds the full story of when, and how many, solutions exist. To test your own instruments and targets, use the **column-space explorer** in the Examples tab.

IN ONE LINE

$C(A)$ = everything the columns can build = the targets your funds can reach; each new column either adds a direction or adds nothing.

▼ Check yourself (Strang's example): what is the column space of $\begin{bmatrix} 1 & 3 \\ 2 & 3 \\ 4 & 1 \end{bmatrix}$? Is $(4, 5, 5)$ in it?

The two columns $(1, 2, 4)$ and $(3, 3, 1)$ point in different directions, so their mixes fill a plane through the origin. And $(4, 5, 5) = 1 \times (1, 2, 4) + 1 \times (3, 3, 1)$, so yes, it's in the plane: it is simply the sum of the two columns.